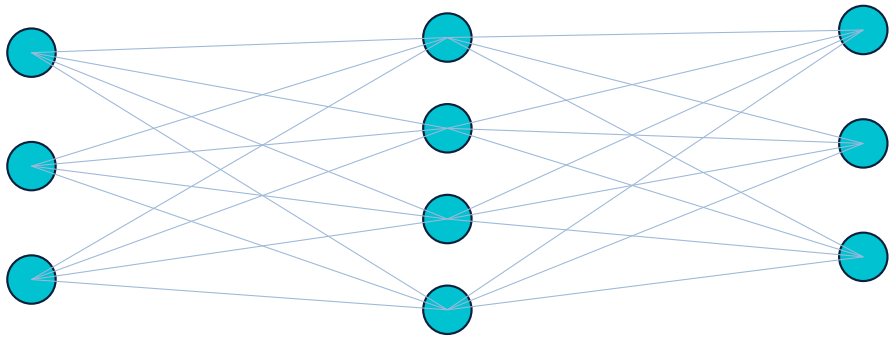
AI BASICS

GRADE 8 • MONTH 1

Chapter 1: How AI Makes Choices

Learning from Data

Data inputs → AI model (pattern finder) → Prediction / choice



What you will learn in this workbook

- What Artificial Intelligence (AI) is and where we use it every day.
- What data, datasets, labels, features and patterns mean.
- How AI learns from examples using machine learning.
- How AI uses training data to make predictions and how we check accuracy.
- How classification, recommendations and prediction errors work.
- Why data quality, bias, fairness, privacy and responsible AI matter.

Student details

Name: _____ Class: _____ Roll No.: _____

School: _____ Teacher: _____

Contents

Your learning journey for Month 1

Section	What it contains
1. Introduction to AI	Meaning of AI and everyday applications
2. Data and Patterns	Data, datasets, labels, features and patterns
3. Machine Learning	Training data, testing data, models and evaluation
4. How AI Makes Choices	Classification, recommendations and prediction errors
5. Fruit Classifier Example	A complete worked example with a small dataset
6. Accuracy and Responsible AI	Accuracy, data quality, bias, fairness and privacy
7. AI in the Real World	Education, healthcare, agriculture, transport, environment and marine science
8. Summary and Glossary	Chapter summary, concept map and key terms
9. Classroom Activities	Five hands-on activities (no computer needed)
10. Worksheets	Three practice worksheets
11. MCQs, Quiz and Assessment	25 MCQs, a 10-question quiz and a 30-mark assessment
12. Puzzles and Brain Games	Word search, matching, patterns and decision detective
13. Projects and Reflection	Case studies, mini-project, HOT questions, revision and reflection
14. Teacher Answer Key	Answers, explanations and marking guidelines

How to use this workbook

Read each section carefully. Complete the activities and worksheets in the spaces provided. Use the glossary whenever you meet a new word. Your teacher will use the answer key at the end for checking and discussion.

1. Introduction to AI

What is Artificial Intelligence?

1.1 What is AI?

Artificial Intelligence (AI) is a branch of computer science that creates computer systems which can perform tasks that usually need human intelligence. These tasks include recognising faces, understanding speech, translating languages, recommending videos and identifying objects in photos.

AI does not “think” exactly like a person. It follows mathematical patterns found in large amounts of data. For example, an AI system can learn that emails containing certain words are often spam, and then use that pattern to filter new emails.

Key idea
AI learns patterns from examples (data) and uses those patterns to make useful predictions or decisions.

1.2 AI in everyday life

Where you see AI	What the AI does	Data it may learn from
Smartphone camera	Recognises faces and improves photos	Thousands of labelled photos
Video apps	Recommends videos you may enjoy	Your watching history and similar users’ choices
Maps and navigation	Suggests the fastest route	Road conditions, speed and past journey times
Voice assistants	Understands spoken questions	Large collections of recorded speech and text
Online shopping	Suggests products	Past purchases, searches and product information
Email	Filters spam	Examples of spam and non-spam emails

1.3 AI is not magic

AI systems are powerful, but they have limits. They can make mistakes, especially when the data they learned from is incomplete, incorrect or unfair. Humans must check important AI decisions, especially in healthcare, education, transport and safety.

Think about it
Name one AI tool you have used this week. What data do you think it used to help you? Write your answer in your notebook or in the reflection page at the end of this workbook.

2. Understanding Data and Patterns

The fuel that powers AI

2.1 What is data?

Data is information collected about people, things, events or places. Data can be numbers, words, pictures, sounds, dates or sensor readings. AI needs data because it learns by finding regularities, called **patterns**, in examples.

Example: A weather app collects data such as temperature, humidity, wind speed and rainfall. From past data, it can estimate tomorrow's weather.

2.2 Datasets

A **dataset** is an organised collection of data, usually arranged like a table. Each row is one example, and each column is one type of information.

Student	Hours studied	Sleep (hours)	Attendance (%)	Result
Asha	4	8	95	Pass
Ravi	1	5	70	Fail
Meera	3	7	90	Pass
Arjun	2	6	80	Pass

In this small dataset, each student is one example. The AI may notice that students with more study hours and higher attendance are more likely to pass. This is a pattern—but it is only a pattern, not a guarantee.

2.3 Features and labels

A **feature** is a piece of information used to make a decision. A **label** is the answer or category we want the AI to learn.

Task	Features (clues)	Label (answer)
Identifying an animal	Number of legs, body covering, sound, size	Cat, dog, cow or bird
Spam detection	Words used, sender, number of links	Spam or not spam
Fruit classification	Colour, shape, size, weight	Apple, banana or lime
Predicting exam result	Study hours, attendance, sleep	Pass or fail

2.4 Patterns

A **pattern** is a repeated relationship in data. AI searches for patterns such as "red and round fruits are often apples" or "messages with many unknown links are often spam."

Remember
Features are the clues. Labels are the answers. Patterns connect the clues to the answers.

3. How AI Learns: Machine Learning

From examples to predictions

3.1 What is machine learning?

Machine learning (ML) is a method used by AI systems to learn rules and patterns from data instead of being programmed with every rule by hand. The more good-quality examples the system sees, the better it can become at recognising patterns.

For example, instead of writing rules such as “if a fruit is red and round, call it an apple,” we can show the computer many labelled fruit photos. The machine learning system finds useful patterns by itself.

3.2 The machine-learning process

Collect data	Prepare and label data	Train the model	Test the model	Use it for predictions

Step	What happens	Fruit example
1. Collect data	Gather examples	Collect photos and details of apples, bananas and limes
2. Label data	Add the correct answer to each example	Mark each photo as apple, banana or lime
3. Train	Let AI find patterns in most of the data	AI learns that yellow + long often means banana
4. Test	Check the AI on new examples it has not seen	Give it a new fruit photo and compare its answer with the truth
5. Predict	Use the trained AI in real life	Identify fruits in a market or storage area

3.3 Training data and testing data

Training data is the set of examples an AI uses to learn. **Testing data** is a separate set of examples used to check how well the AI performs on new data. We never let the AI see the testing answers while it is learning.

Why separate the data?
If we test an AI using the same examples it learned from, it may look perfect simply because it has memorised those examples. Testing with new examples shows whether it has truly learned a useful pattern.

3.4 AI models

An **AI model** is the trained program that stores the patterns learned from data. Think of it as a “pattern-finder.” When we give it new features, it gives an output called a **prediction**.

New features	Trained AI model	Prediction

Example: Features: yellow, long, medium size → Model → Prediction: banana.

3.5 Evaluating an AI model

Evaluation means checking how often the AI’s predictions are correct. A common measure is **accuracy**.

Accuracy formula
Accuracy = (Number of correct predictions ÷ Total number of predictions) × 100

Worked example: An AI classifies 20 fruits. It gets 17 correct and 3 wrong. Accuracy = 17 ÷ 20 × 100 = 85%.

4. How AI Makes Choices

Classification, recommendations and errors

4.1 Classification

Classification is when AI places something into a category. The AI looks at features and chooses the most likely label.

Example	Categories	Features used
Email filter	Spam / Not spam	Words, links, sender
Plant leaf checker	Healthy / Diseased	Colour, spots, shape
Fruit sorter	Apple / Banana / Lime	Colour, shape, size
Attendance system	Present / Absent	Face image or ID

4.2 Recommendations

A **recommendation system** suggests items a person may like. It uses patterns from your past choices and from people with similar choices. For example, if you often watch science videos, the system may recommend more science videos.

Recommendations are not always “right” or “wrong” in the same way as classification. Instead, we ask whether they are useful, relevant and fair.

4.3 Prediction errors

AI can make two main kinds of mistakes:

- **False positive:** The AI says “yes” when the correct answer is “no.” Example: an email filter marks a normal message as spam.
- **False negative:** The AI says “no” when the correct answer is “yes.” Example: a spam filter misses an actual spam email.
- **Wrong category:** In classification, the AI chooses the incorrect label. Example: it calls a lime an apple.

4.4 Why errors happen

- The training data was too small.
- The examples were unclear or labelled incorrectly.
- The new example is very different from the training data.
- Important features were missing.
- The data was biased or unbalanced.

Important

A high accuracy score does not always mean an AI is fair or safe. We must also check who is included in the data, what mistakes it makes, and who could be harmed by those mistakes.

5. Worked Example: Classifying Fruits

A complete AI decision system

Let us build a very small fruit-classification AI. It will decide whether a fruit is an **apple**, **banana** or **lime** using colour, shape and size.



Apple: red, round



Banana: yellow, long



Lime: green, small

Features (colour, shape, size) help the AI decide the label (fruit name).

5.1 Our dataset

Fruit No.	Colour	Shape	Size	Correct label
1	Red	Round	Medium	Apple
2	Red	Round	Medium	Apple
3	Green	Round	Small	Lime
4	Green	Round	Small	Lime
5	Yellow	Long	Medium	Banana
6	Yellow	Long	Medium	Banana
7	Red	Round	Small	Apple
8	Yellow	Round	Small	Lime
9	Green	Long	Medium	Banana
10	Red	Round	Medium	Apple

Here, **colour, shape and size** are features. The fruit name is the **label**.

5.2 Patterns the AI may learn

Pattern	Likely prediction
Red + round	Apple
Yellow + long	Banana
Green + round + small	Lime
Yellow + round + small	Lime (in this dataset)
Green + long	Banana (in this dataset)

5.3 Testing the fruit AI

Suppose we test the trained AI with five new fruits:

Test fruit	Features	AI prediction	Correct label	Correct?
T1	Red, round, medium	Apple	Apple	Yes
T2	Yellow, long, medium	Banana	Banana	Yes
T3	Green, round, small	Lime	Lime	Yes
T4	Purple, round, medium	Apple	Plum	No
T5	Yellow, round, small	Lime	Lemon	No

The AI got 3 correct answers out of 5. Accuracy = 3 ÷ 5 × 100 = **60%**.

5.4 What went wrong?

- The AI had never seen purple fruits, so it guessed the closest learned pattern.
- It had no “lemon” category, so it could not give the correct label.
- The dataset was very small, so some patterns were incomplete.

Lesson
AI can only choose from the categories it was trained to know. If a new type of fruit appears, the system may make a wrong prediction unless humans update and improve the data.

6. Accuracy, Data Quality and Responsible AI

Making AI helpful, fair and safe

6.1 Accuracy and incorrect predictions

Accuracy tells us the percentage of correct predictions. But accuracy alone is not enough. We should also look at **which** examples the AI gets wrong.

For example, a fruit AI may be 90% accurate overall but always misidentifies green apples. That specific error could cause problems in a fruit-sorting factory.

6.2 Data quality

Good-quality data	Poor-quality data
Correct labels	Wrong or missing labels
Represents many kinds of examples	Contains only one type of example
Collected with permission	Collected without consent
Checked for errors	Contains duplicates and mistakes
Updated when the world changes	Old and outdated

6.3 Bias and fairness

Bias is an unfair pattern in an AI system, often caused by unbalanced or incomplete data. If a face-recognition system is trained mostly on photos of people from one region, it may work poorly for people from other regions.

Fairness means the AI should work reasonably well for different groups of people and should not unfairly disadvantage anyone.

6.4 Privacy

Privacy means protecting personal information. AI often uses data about people, so we must collect only necessary data, keep it secure, and ask permission where required.

6.5 Responsible AI

- Use accurate, diverse and permission-based data.
- Test AI with many different groups and situations.
- Keep humans involved in important decisions.
- Explain AI decisions clearly when possible.
- Protect personal and private information.
- Correct errors and improve the system regularly.

Golden rule
AI should assist humans—not replace human judgement, care and responsibility.

7. AI in the Real World

Applications across subjects and careers

Field	How AI helps	Example data used
Education	Personalised practice, checking answers, identifying students who need support	Quiz scores, learning time, question types
Healthcare	Helps doctors examine X-rays, predict risks and manage hospital records	Medical images, symptoms, test results
Agriculture	Detects crop diseases, predicts weather effects and supports irrigation	Drone images, soil data, rainfall records
Transportation	Plans routes, manages traffic and supports driver-assistance systems	GPS data, traffic flow, camera images
Environmental protection	Predicts floods, tracks pollution and monitors forests	Satellite images, sensor readings, weather data
Marine science	Tracks fish populations, identifies species and monitors ocean health	Underwater images, sound recordings, temperature data

7.1 Case Study 1: AI in agriculture

A farmer notices yellow spots on crop leaves. A smartphone AI app analyses a photo and suggests a possible fungal disease. The farmer uses the suggestion as advice, checks with an agricultural officer, and then decides the treatment.

Discussion
What features might the AI use? Why should the farmer not depend on the app alone? What could happen if the app was trained mostly on crops from a different region?

7.2 Case Study 2: AI in marine science

Researchers place underwater cameras near a coral reef. An AI model learns to identify fish species from thousands of labelled images. It helps scientists count fish and notice changes in the reef ecosystem faster than manual checking alone.

Discussion
Why is a large and varied image dataset important here? How could poor water visibility cause prediction errors?

8. Chapter Summary and Concept Map

Revise the big ideas

8.1 Chapter summary

- AI is computer technology that performs tasks requiring human-like intelligence, such as recognition and prediction.
- Data is information; a dataset is an organised collection of examples.
- Features are clues; labels are correct answers; patterns connect them.
- Machine learning trains an AI model using examples.
- Training data teaches the model; testing data checks it on new examples.
- Classification puts items into categories; recommendation systems suggest items.
- Prediction errors include false positives, false negatives and wrong categories.
- Accuracy measures the percentage of correct predictions.
- Good data should be accurate, diverse, permission-based and up to date.
- Responsible AI protects privacy, reduces bias and keeps humans in charge.

8.2 Concept map

Data	Features + labels	Patterns	AI model	Prediction	Check accuracy & fairness

If any step uses poor-quality or biased data, the final prediction may be unfair or incorrect.

8.3 Glossary

Term	Meaning
Artificial Intelligence (AI)	Computer systems that perform tasks needing human-like intelligence
Data	Information such as numbers, words, images or sounds
Dataset	An organised collection of data examples
Feature	A clue or characteristic used to make a decision
Label	The correct answer or category for an example
Pattern	A repeated relationship found in data
Machine Learning	A method where AI learns patterns from data
Training data	Examples used to teach an AI model
Testing data	New examples used to check the model
AI model	The trained system that makes predictions
Prediction	The AI’s best guess or output
Classification	Placing items into categories
Recommendation	A suggestion based on patterns in data
Accuracy	Percentage of correct predictions

Term	Meaning
Bias	Unfair pattern caused by unbalanced data
Fairness	Treating different groups reasonably and without unfair disadvantage
Privacy	Protecting personal information
Responsible AI	Using AI safely, fairly, transparently and ethically

9. Classroom Activities

Learn by doing—no computer required

Activity 1: Human vs AI Classification

Objective
Classify everyday items using human reasoning and compare it with how an AI classifier works.

Materials: Paper cards, pen, table below.

Steps:

- Your teacher will give each group 12 cards with items such as “mango,” “car,” “rose,” “dog,” “bus,” and “tulip.”
- Sort the cards into two groups: Living things and Non-living things.
- Write the feature you used for each card, such as “grows,” “moves by itself,” or “made by humans.”
- Now imagine an AI trained only on pictures. Which features could it use?
- Discuss one item that may confuse the AI and explain why.

Student task: Complete the table.

Item	Group	Feature you used	Could AI get confused? Why?

Expected outcome: Students understand that both humans and AI use features, but humans also use context, experience and common sense.

Discussion questions: What features were most useful? When can an AI make a mistake even with clear features?

Teacher guidance: Encourage students to justify every grouping with a feature. Highlight that AI does not “know” meanings; it detects statistical patterns.

Activity 2: Build a Mini Dataset

Objective
Create a small labelled dataset and identify features and patterns.

Materials: Notebook, ruler, coloured pencils.

Steps:

- Choose a topic: school bags, leaves, shoes, or tiffin boxes.
- Collect information about 8 examples. Choose at least three features.
- Add a label for each example.
- Write two patterns you notice.
- State one example that does not fit the pattern.

Student task: Complete your dataset.

Example	Feature 1	Feature 2	Feature 3	Label

Pattern 1: _____

Pattern 2: _____

Example that does not fit: _____

Expected outcome: Students can distinguish features from labels and recognise simple patterns.

Teacher guidance: Ask groups to swap datasets and find one additional pattern. Remind them that a pattern from a small dataset may not always be true.

Activity 3: Train a Classroom Classifier

Objective
Simulate how an AI learns rules from training examples and predicts labels for new examples.

Materials: Paper slips, two boxes labelled “Training” and “Testing.”

Steps:

- Create 10 paper slips describing animals using only three features: number of legs, body covering, and sound.
- Write the correct animal name on the back of each slip as its label.
- Put 7 slips in the Training box and 3 in the Testing box.
- As a class, read the training slips and write rules such as “four legs + fur + barking = dog.”
- Use the rules to predict labels for the testing slips.
- Turn over the slips and check the answers.

Student task: Write two classification rules your class discovered.

Rule 1: _____

Rule 2: _____

Expected outcome: Students experience training, testing, prediction and evaluation without a computer.

Discussion questions: Which rule was most reliable? What would happen if one training label was wrong?

Teacher guidance: Use this to introduce overfitting informally: rules that fit only the training examples may fail on new examples.

Activity 4: Identify Bias in a Dataset

Objective
Recognise how unbalanced data can create unfair AI predictions.

Materials: Printed dataset below, pencils.

Photo number	Type of dog	Image quality	Label used in training
1	Labrador	Clear	Dog
2	Labrador	Clear	Dog
3	Labrador	Clear	Dog
4	Labrador	Clear	Dog
5	Beagle	Blurry	Dog
6	Beagle	Blurry	Dog
7	Beagle	Blurry	Dog
8	Local street dog	Very blurry	Dog

Steps:

- Count how many examples belong to each type of dog.
- Identify which type has the fewest clear examples.
- Predict what may happen when the AI sees a clear photo of a street dog.
- Suggest two ways to improve the dataset.

Student task: Write your answers.

1. Which type of dog has the fewest clear training examples? _____

2. What prediction error might occur? _____

3. Two improvements: _____

Expected outcome: Students explain that unequal representation can reduce fairness and accuracy for less-represented groups.

Teacher guidance: Connect this to real-world facial recognition and speech recognition bias. Emphasise that bias is a data and design problem, not a problem with the people represented.

Activity 5: Draw an AI Decision-Making Flowchart

Objective
Show the steps an AI follows from input data to a decision.

Scenario: An AI helps a school library decide whether a book is suitable for Grade 8 students. It examines reading level, topic and content warnings.

Steps:

- Draw a start box: “New book arrives.”
- Add feature boxes: reading level, topic, content warnings.
- Add a decision box: “Does the book match Grade 8 reading level?”
- Add two paths: Yes → Recommend; No → Suggest another book.
- Add a final box: “Human librarian reviews the decision.”

Student task: Draw your flowchart in the space below. Use arrows and label every box.

Expected outcome: A labelled flowchart showing data input, feature checking, decision, output and human review.

Teacher guidance: Accept any clear sequence that includes features, a decision point, an output and human oversight.

10. Worksheets

Practice what you have learned

Worksheet 1: AI and Data Basics

A. Fill in the blanks.

1. _____ is information collected about people, things or events.
2. An organised collection of data is called a _____.
3. A _____ is a clue or characteristic used to make a decision.
4. The correct answer or category for an example is called a _____.
5. A repeated relationship in data is called a _____.

B. Match the following.

Column A	Column B
1. AI	a. Learns patterns from examples
2. Dataset	b. A computer system that performs intelligent tasks
3. Feature	c. Organised collection of examples
4. Machine learning	d. A clue used for decision-making

C. True or False.

1. AI always makes perfect decisions. (_____)
2. Data can include images and sounds. (_____)
3. A label is the answer we want AI to learn. (_____)
4. AI can learn without any data. (_____)
5. Patterns help AI make predictions. (_____)

D. Short answer.

6. Define artificial intelligence in your own words.

7. Give two examples of AI applications used in daily life.

8. What is the difference between a feature and a label?

9. Look at this data: “Red, round, medium size → Apple.” Identify the features and the label.

10. Why does AI need data?

11. Name three types of data that can be used to train AI.

12. Give one example of a pattern in your school life.

13. What is a dataset? Give one example.

14. Why should a dataset contain many different examples?

15. Scenario: A music app recommends songs. What data might it use?

Worksheet 2: Learning from Data and Making Predictions

A. Fill in the blanks.

1. _____ data is used to teach an AI model.
2. _____ data is used to check the model on new examples.
3. The trained AI system that makes predictions is called an AI _____.
4. _____ means placing items into categories.
5. _____ = (correct predictions ÷ total predictions) × 100.

B. True or False.

1. Testing data should be different from training data. (_____)
2. A prediction is always correct. (_____)
3. Machine learning finds patterns from examples. (_____)
4. A recommendation system suggests items based on patterns. (_____)
5. More data always guarantees perfect accuracy. (_____)

C. Data interpretation.

An AI classified 25 emails. It correctly identified 21 emails.

6. Calculate the accuracy. Show your working.

7. How many emails did it classify incorrectly?

8. Suggest one possible reason for the incorrect predictions.

D. Short answer and application.

9. What is the difference between training data and testing data?

10. Explain classification with one real-life example.

11. What is a recommendation system? Give one example.

12. A fruit AI sees a purple, round fruit but has only been trained on apples, bananas and limes. What may happen? Why?

13. Write the machine-learning process in the correct order: Test, Collect data, Train, Predict, Label data.

14. Why do we evaluate an AI model?

15. Scenario: A school wants AI to predict which students may need extra maths practice. Suggest three useful features.

Worksheet 3: AI Errors, Bias, and Responsible AI

A. Fill in the blanks.

1. A _____ positive occurs when AI says “yes” but the correct answer is “no.”
2. A _____ negative occurs when AI says “no” but the correct answer is “yes.”
3. _____ is an unfair pattern in an AI system.
4. _____ means protecting personal information.
5. Responsible AI keeps _____ involved in important decisions.

B. Match the following.

Column A	Column B
1. Bias	a. Percentage of correct predictions
2. Privacy	b. Unfair pattern in AI
3. Accuracy	c. Protecting personal data
4. False negative	d. AI misses something that is true

C. True or False.

1. Bias can occur when training data is unbalanced. (_____)
2. AI systems should always make final medical decisions without doctors. (_____)
3. Personal data should be collected without permission. (_____)
4. Data quality affects AI predictions. (_____)
5. Responsible AI includes fairness and privacy. (_____)

D. Scenario-based questions.

6. A face-recognition system works well for one group of students but poorly for another. What problem is shown? Suggest one solution.

7. A health AI gives a wrong diagnosis for a patient. Why should a doctor review the result?

8. Give two ways to improve the quality of a dataset.

9. What is the difference between a false positive and a false negative? Give one example of each.

10. Why is diverse data important for fairness?

11. A school AI uses students' private health information without permission. Which responsible-AI principle is violated?

12. Give two examples of how humans should supervise AI decisions.

13. Why can a highly accurate AI still be unfair?

14. A crop-disease AI was trained only on images from one district. What limitation may occur in another district?

15. Write two rules for using AI responsibly at school.

11. MCQs, Quiz and Assessment

Check your understanding

Multiple-Choice Questions (25 marks)

Choose the correct option. Each question carries 1 mark.

1. AI mainly learns to make decisions from —

(a) Random guesses (b) Data and patterns (c) Human emotions only (d) Internet speed

2. A collection of organised examples is called a —

(a) Feature (b) Label (c) Dataset (d) Prediction

3. The correct answer assigned to an example is its —

(a) Feature (b) Label (c) Model (d) Pattern

4. Colour, shape and size of a fruit are examples of —

(a) Labels (b) Features (c) Errors (d) Models

5. Machine learning means —

(a) Copying human brains exactly (b) Learning patterns from data (c) Typing faster (d) Storing files

6. Data used to teach an AI model is called —

(a) Testing data (b) Training data (c) Random data (d) Private data

7. Data used to check an AI model is called —

(a) Training data (b) Testing data (c) Old data (d) Labelled data only

8. An AI model gives an output called a —

(a) Dataset (b) Feature (c) Prediction (d) Pattern

9. Placing emails into “spam” or “not spam” is —

(a) Recommendation (b) Classification (c) Privacy (d) Bias

10. Suggesting a video you may like is an example of —

(a) Classification (b) Recommendation (c) Testing (d) Labelling

11. Accuracy is calculated as —

(a) Wrong ÷ total × 100 (b) Correct ÷ total × 100 (c) Total ÷ correct × 100 (d) Correct × wrong

12. An AI says an email is spam when it is not. This is a —

(a) True positive (b) False positive (c) False negative (d) Correct prediction

13. An AI misses an actual spam email. This is a —

(a) False positive (b) False negative (c) True negative (d) Classification

14. A model gets 18 out of 20 predictions correct. Its accuracy is —

(a) 80% (b) 85% (c) 90% (d) 95%

15. Poor data quality can cause —

(a) Better predictions (b) More errors (c) More privacy (d) Less data

16. Bias in AI often comes from —

(a) Balanced data (b) Unbalanced or incomplete data (c) Too many correct labels (d) Fast computers

17. Fairness in AI means —

(a) One group always wins (b) AI works reasonably for different groups (c) Data is secret (d) Predictions are random

18. Privacy means —

(a) Sharing all personal data (b) Protecting personal information (c) Deleting all AI models (d) Ignoring permissions

19. Responsible AI requires —

(a) Humans checking important decisions (b) No testing (c) Secret data collection (d) Ignoring errors

20. Which is a feature for predicting exam results?

(a) Student's name only (b) Study hours (c) Random number (d) School building colour

21. A fruit AI trained only on red apples may wrongly classify —

(a) Red apples (b) Green apples (c) Round red fruits (d) All apples

22. Which field uses AI to detect crop diseases?

(a) Marine science (b) Agriculture (c) Transportation (d) Education

23. AI in healthcare should be supervised by —

(a) Advertisements (b) Doctors and experts (c) Random users (d) No one

24. Which is an example of personal data?

(a) Favourite colour of a fictional robot (b) A student's health record (c) The number 7 (d) A public map

25. The best first step before training an AI is —

(a) Collect and prepare relevant data (b) Deploy it immediately (c) Delete all labels (d) Ignore testing

Quick Quiz: 10 Questions

Answer in one word or one sentence.

1. What does AI stand for?

2. What is a dataset?

3. What are features?

4. What is a label?

5. What is machine learning?

6. What is training data?

7. What is a prediction?

8. What is classification?

9. What is accuracy?

10. Name one principle of responsible AI.

Chapter Assessment: 30 Marks

Section A: Multiple Choice (1 mark each = 5 marks)

1. AI learns mainly from: (a) data (b) luck (c) colours only (d) nothing

2. A feature is: (a) the answer (b) a clue (c) an error (d) a dataset

3. Testing data is used to: (a) train the AI (b) check the AI (c) delete data (d) label data

4. Classification means: (a) suggesting items (b) placing into categories (c) deleting data (d) drawing charts

5. Bias means: (a) fairness (b) an unfair pattern (c) privacy (d) accuracy

Section B: Short Answer (2 marks each = 10 marks)

1. Define machine learning.

2. Differentiate training data and testing data.

3. What is a prediction? Give one example.

4. Explain classification with an example.

5. What is data quality? Why is it important?

Section C: Application (3 marks each = 9 marks)

1. An AI classifies 40 fruits and gets 34 correct. Calculate its accuracy. Show your working.

2. A video app recommends only one type of video to a student. Explain one possible limitation and one improvement.

3. A hospital AI helps detect pneumonia from X-rays. Explain why doctors must review its predictions.

Section D: Higher-Order Thinking (6 marks)

1. Design a small AI system to identify whether a plant leaf is healthy or diseased. State: (a) two features, (b) the label, (c) one possible error, and (d) one ethical concern. (4 marks)

2. “More data always makes AI fair.” Do you agree? Justify your answer. (2 marks)

Total: 5 + 10 + 9 + 6 = 30 marks

12. Puzzles and Brain Games

AI learning can be fun

Puzzle 1: AI and Data Word Search

Find these words: DATA, LABEL, MODEL, PATTERN, TRAINING, TESTING, BIAS, PRIVACY, FEATURE, CLASSIFY. Words may go across or down.

A	T	R	A	I	N	I	N	G	X
D	A	T	A	B	M	O	D	E	L
P	R	I	V	A	C	Y	F	A	I
T	E	S	T	I	N	G	T	U	R
E	L	A	B	E	L	R	U	E	A
R	N	C	L	A	S	S	I	F	Y
N	I	O	P	A	T	T	E	R	N
B	I	A	S	D	F	E	A	T	U
M	O	D	E	L	P	A	T	T	E
Y	C	A	V	A	R	P	I	V	D

Solution: DATA (row 2), LABEL (row 5), MODEL (row 2 and row 10), PATTERN (row 7), TRAINING (row 1), TESTING (row 4), BIAS (row 8), PRIVACY (row 3), FEATURE (row 8), CLASSIFY (row 6).

Puzzle 2: Match AI Terms with Their Meanings

Write the correct letter beside each term.

Term	Meanings
1. Dataset	A. Unfair pattern in AI
2. Feature	B. Trained system that predicts
3. Label	C. Organised collection of examples
4. Model	D. Clue used for a decision
5. Bias	E. Correct answer or category
6. Accuracy	F. Percentage of correct predictions

Solution: 1-C, 2-D, 3-E, 4-B, 5-A, 6-F.

Puzzle 3: Find the Pattern Challenge

Study the sequence and write the next item and the rule.

Apple (red, round) → Banana (yellow, long) → Lime (green, round) → Apple (red, round) → Banana (yellow, long) → _____

Rule: _____

Solution: Lime (green, round). The sequence repeats apple → banana → lime.

Puzzle 4: AI Decision Detective

A library AI recommends books using reading level and topic. Use the clues to decide which book each student receives.

Student	Reading level	Interest	AI recommendation
Riya	Level 4	Space	?
Omar	Level 2	Animals	?
Sara	Level 5	Mystery	?

Clues: Level 4 + Space = “Planet Quest”; Level 2 + Animals = “Animal Friends”; Level 5 + Mystery = “The Secret Lighthouse.”

Solution: Riya—Planet Quest; Omar—Animal Friends; Sara—The Secret Lighthouse.

13. Projects, Reflection and Revision

Think like an AI designer

Mini-Project: Design a Simple AI Decision System

Choose one problem in your school or community. Design an AI decision system on paper. No programming is required.

Project step	Your response
1. Problem	
2. Who will use it?	
3. Data to collect	
4. Features	
5. Labels / outputs	
6. Pattern the AI may learn	
7. AI prediction or recommendation	
8. One limitation	
9. One ethical concern	
10. How humans will check decisions	

Higher-Order Thinking Questions

1. Can an AI be accurate but unfair? Explain with an example.

2. Why might an AI trained in one country fail in another country?

3. If an AI recommends videos that keep people watching for hours, is it always making a good decision? Explain.

4. How can students help make school AI systems more responsible?

Revision Exercises

1. Write five AI terms and their meanings.

2. Draw the machine-learning process as a flowchart.

3. Give one example each of AI in healthcare, agriculture and marine science.

Student Reflection

1. Three new things I learned:

2. One question I still have:

3. One way I can use AI responsibly:

Self-Assessment Checklist

I can…	Yes	Not yet
Explain what AI is	■	■
Identify data, features and labels	■	■
Describe machine learning	■	■
Explain training and testing data	■	■
Calculate simple accuracy	■	■
Explain classification and recommendations	■	■
Identify bias and privacy issues	■	■
Give real-life AI examples	■	■

14. Teacher Answer Key

For teacher use only

Worksheet 1: AI and Data Basics

A. 1. Data 2. Dataset 3. Feature 4. Label 5. Pattern.

B. 1-b, 2-c, 3-d, 4-a.

C. 1. False 2. True 3. True 4. False 5. True.

D. 6. AI is computer technology that performs tasks needing human-like intelligence, such as recognising images or making predictions. 7. Any two: smartphone face unlock, maps, voice assistants, video recommendations, spam filters. 8. A feature is a clue; a label is the correct answer/category. 9. Features: red, round, medium size; label: Apple. 10. AI needs data to find patterns and make predictions. 11. Numbers, text/words, images, audio, sensor readings (any three). 12. Accept any valid repeated school-life relationship. 13. An organised collection of examples; e.g., a table of students' study hours and results. 14. Diverse examples help AI learn fair and general patterns. 15. Songs listened to, searches, liked songs, time of day, similar listeners' choices.

Worksheet 2: Learning from Data and Making Predictions

A. 1. Training 2. Testing 3. Model 4. Classification 5. Accuracy.

B. 1. True 2. False 3. True 4. True 5. False.

C. 6. 21 ÷ 25 × 100 = 84%. 7. 4 emails. 8. Accept: unclear words, misleading links, insufficient examples, wrong labels, new spam style.

D. 9. Training data teaches the model; testing data checks it on unseen examples. 10. Classification places items into categories; e.g., spam/not spam. 11. It suggests items based on patterns; e.g., video recommendations. 12. It may make a wrong prediction because purple fruits were not in its training categories. 13. Collect data → Label data → Train → Test → Predict. 14. To measure performance and find errors before real use. 15. Any three: previous maths scores, homework completion, class participation, attendance, practice-test results.

Worksheet 3: AI Errors, Bias, and Responsible AI

A. 1. False 2. False 3. Bias 4. Privacy 5. Humans.

B. 1-b, 2-c, 3-a, 4-d.

C. 1. True 2. False 3. False 4. True 5. True.

D. 6. Bias/unfair performance; use more diverse and representative training images. 7. Doctors provide medical judgement, check context and prevent harm from AI errors. 8. Correct labels, remove duplicates/errors, include diverse examples, update data. 9. False positive: AI says yes when answer is no; false negative: AI says no when answer is yes; examples may vary. 10. It helps the AI work reasonably for different groups. 11. Privacy/consent. 12. Accept: teacher reviews recommendations; doctor reviews diagnosis; librarian reviews book suitability; driver remains alert. 13. It may perform well overall but fail for a smaller or under-represented group. 14. It may not recognise local crop varieties, diseases, lighting or soil conditions. 15. Accept any two responsible-use rules.

MCQ Answer Key

1. (b)
2. (c)
3. (b)
4. (b)
5. (b)
6. (b)
7. (b)
8. (c)
9. (b)
10. (b)
11. (b)
12. (b)
13. (b)
14. (c)
15. (b)
16. (b)
17. (b)
18. (b)
19. (a)
20. (b)
21. (b)
22. (b)
23. (b)
24. (b)
25. (a)

Marking: 1 mark for each correct answer; total 25 marks.

Quick Quiz Answers

1. Artificial Intelligence. 2. An organised collection of data examples. 3. Clues or characteristics used for decisions. 4. The correct answer/category. 5. Learning patterns from data. 6. Examples used to teach the model. 7. The AI's output or best guess. 8. Placing items into categories. 9. Percentage of correct predictions. 10. Fairness, privacy, transparency, human oversight, or data quality.

Assessment Marking Scheme

Section A: 1-a, 2-b, 3-b, 4-b, 5-b. (5 marks)

Section B: Award 2 marks for a correct definition/example with clear explanation; 1 mark for partial understanding.

1. Machine learning is a method where AI learns patterns from data instead of following only hand-written rules.

2. Training data teaches the model; testing data checks performance on new, unseen examples.

3. A prediction is the AI's output; e.g., predicting "banana" for a yellow, long fruit.

4. Classification places items into categories; e.g., spam or not spam.

5. Data quality means data being accurate, complete, relevant and representative; poor quality causes errors.

Section C: 1. 34 ÷ 40 × 100 = 85%. (3 marks: formula 1, working 1, answer 1)

2. Limitation: narrow recommendations may reduce exposure to new topics; improvement: include varied categories and allow user feedback. (3 marks)

3. Doctors verify context, image quality and patient history; AI errors could affect treatment; human oversight improves safety. (3 marks)

Section D: 1. Features: leaf colour, spots, shape; label: healthy/diseased; error: false negative—diseased leaf marked healthy; ethical concern: wrong advice may harm crops or farmers. (4 marks)

2. Disagree. More data helps only if it is accurate, relevant, diverse and representative; biased data can remain unfair even in large amounts. (2 marks)

Total: 30 marks.

Puzzle Solutions

Puzzle 1: See word locations listed beneath the grid. Puzzle 2: 1-C, 2-D, 3-E, 4-B, 5-A, 6-F. Puzzle 3: Lime (green, round); repeating sequence. Puzzle 4: Riya—Planet Quest; Omar—Animal Friends; Sara—The Secret Lighthouse.

Activity Expected Outcomes

Activity 1: Students classify using features and recognise AI limitations. Activity 2: Students create labelled datasets and identify patterns. Activity 3: Students simulate training, testing and prediction. Activity 4: Students identify representation bias and propose dataset improvements. Activity 5: Students produce a labelled flowchart with human review.